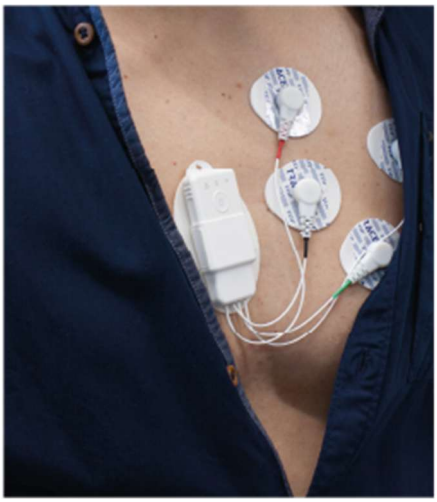
PCARDIO Wearables

User Manual





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1 General safety considerations

We strongly recommend that you follow the safety specifications listed in this section and throughout this manual.

In this user manual, the following conventions are used:

 WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury
 CAUTION	Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury.
NOTICE	Indicates a situation which, if not avoided, could result in damage to property

Important	Indicates significant information which is essential for the product to function correctly
Note	Indicates additional information on usage



➢ This device should not be modified or altered in any way. ➢ The manufacturer is not responsible for any injury or damage caused by using this device for purposes other than those for which it is intended, or by modifications of the device not performed by the manufacturer or an authorized agent.
➢ The equipment should only be operated by qualified personnel.
➢ There are no user serviceable parts inside. Refer servicing to qualified personnel.
➢ Do not use acetone on the recorder. ➢ No part of the device may be autoclaved or sterilized by any means. ➢ Do not immerse any equipment in liquid unless instructions specify that it is safe to do so.

- Only use the supplied components. Do not connect to any other equipment other than specified in this document.
- Do not connect the recorder to any equipment other than that specified in this manual. Only use the cables, sensors or connections provided with this recorder to connect to the recorder.
- When damaged, do not use anymore and return the device for servicing.
- Place the electrodes only on intact skin.
- If any signs of an allergic reaction are observed or you feel irritation while using the device, stop use, remove the sensor and the straps, consult a physician.
- If you feel discomfort while using the system at high ambient temperatures, stop use, remove the strap and consult a physician.
- Do not use the system in a sauna or during swimming.
- Do not use in combination with cardiac pacemakers or other stimulators.
- Do not use the system in combination with high frequency (HF) surgical equipment.
- Before a defibrillator is used on the user, disconnect all devices and electrodes.
- There is a possible hazard, caused by summation of leakage current when several pieces of equipment are interconnected.
- Never connect the product to output carrying voltage. This may cause grievous bodily harm or lead to death.
- Do not connect the electrode strap to any other device while recording.
- Conductive parts of the electrodes should not come into contact with other conductive parts including earth.

Important

- It is advised to perform a short test recording prior to starting any long-term examination as a default Use scenario in order to minimize the risk of malfunctioning of recorder and sensors and amplifiers as for 50/60 Hz humming.

1.1 Safety and Electromagnetic Compatibility

The system has been tested and found to be in compliance with the safety standard IEC 60601-1:2005 + A1:2012+AMD2:2020

- Internally powered equipment during operation, during charging: Class II
- Type BF applied part
- Continuous operation
- Pollution Degree 2
- Over voltage category I environment
- Maximum altitude of 3000m
- Maximum ambient temperature of 45°C

The system has been tested and found to be in compliance with:

- EN 60601-1-2:2015 Ed4
- EN 55011:2016/A1:2017/A11:2020
- EN 300 400 -2:2010
  Test standards: EN 61000-4-2, EN 61000-4-3, EN 61000-4-8.

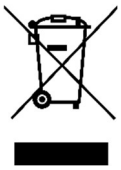
⚠

System is suitable for home healthcare environment and professional healthcare facility environment, except for near active HF surgical equipment and the RF shielded room of a medical electrical system for magnetic resonance imaging.
Use of the device adjacent with other equipment should be avoided because it could result in improper operation. If such use is necessary, this device and the other equipment should be observed to verify that they are operating normally.
Use of equipment and cables other than those provided by the manufacturer could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation.
Portable RF communications equipment should be used no closer than 30 cm to the device. Otherwise, this could result in the degradation of the performance of this device.

1.2 Disposal

The device contains electronic components. In order to avoid risk or harm to the environment through non-professional disposal, the disposal of the device, including all components, must correspond to the valid directives.

The following symbol has been included in accordance with the Waste Electrical and Electronic Equipment (WEEE) Directive of the European Union.



The presence of this marking on the device indicates:
The device was put on the European market after August 13, 2005;
The device is not to be disposed of via the municipal waste collection system of any member state of the European Union.

It is very important that customers understand and follow all laws regarding the proper safe disposal of electrical equipment. For 2M Engineering devices bearing this label, please contact your dealer or the 2M Engineering office for details on the take-back program that will facilitate the proper collection, treatment, recovery, recycling, and safe disposal of the device.

1.3 Explanation of symbols

Symbols used on the Recorder label:

Symbol	Description
	Manufacturer
	Serial number
	Catalogue number
	Date of manufacture
	Operating instructions, consult instructions for use
	Type BF Applied Part
	WEEE symbol (waste electrical and electronic equipment) The symbol indicates that the product should not be discarded as unsorted waste but must be sent to separate collection facilities for recovery and recycling.

2 Device Description

2.1 Intended use

P-CARDIO is intended to measure and record physiological signals for Polygraphy (ECG, bioimpedance) and motion (accelerometer, gyroscope, magnetometer). P-CARDIO is intended to record data for offline review by a researcher.

It does not have ECG analysis software.

Intended population: Humans weighing more than 10kg

P-CARDIO consists of

- Recorder (main unit)
- Recorder holder[1] (electrodes),
- Docking station (for configuration and data download)
- Wireless USB dongle (for visualization and verification of correct functioning).

P-CARDIO (recorder and recorder-holder) can be worn during showering.

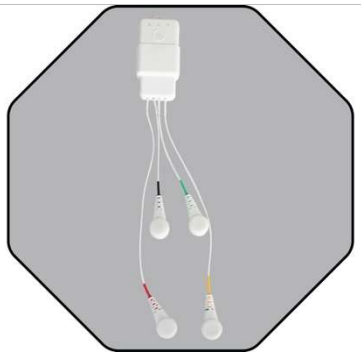
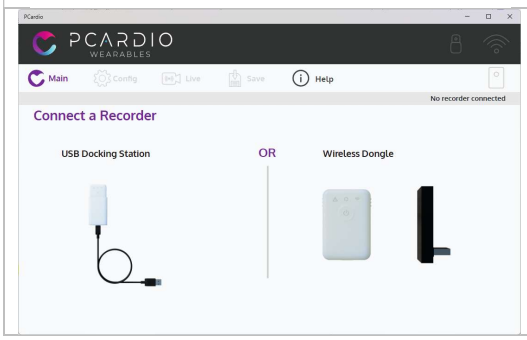
Contraindications

Do not use in combination with cardiac pacemakers or other stimulators. It may interfere with ECG data.
Do not use in combination with a defibrillator. It may damage the device.
Do not use if known to be allergic to silicones, acrylic chemicals or medical gels.

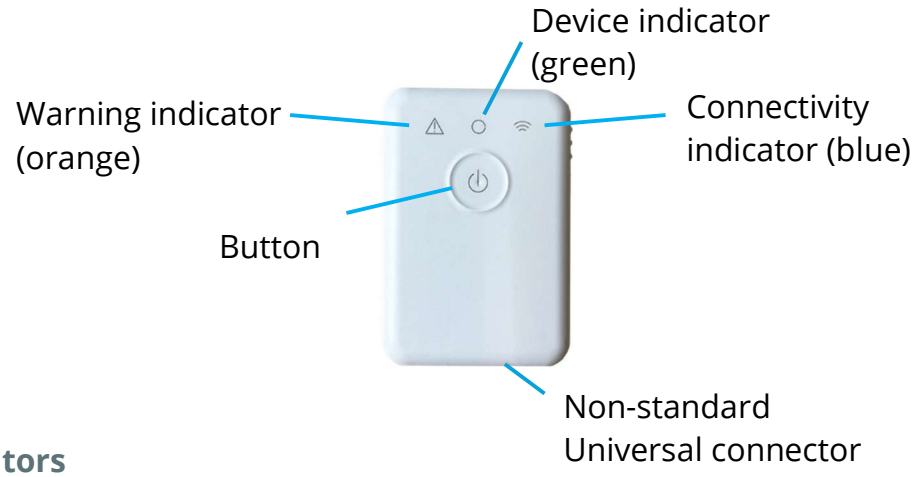
2.2 Device & system overview

	**P-CARDIO Recorder** - **HP201**: 1-Lead ECG + Respiration + Motion - **HP202**: 6-Lead ECG + Motion

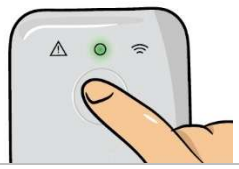
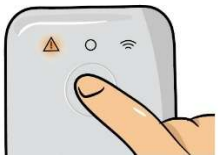
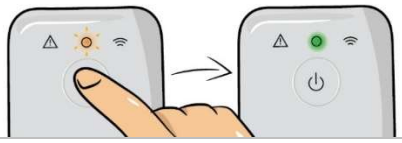
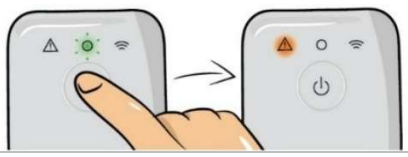
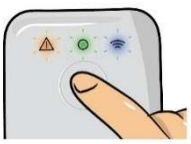
[1] Contact 2M for different types of recorder holders and body attachment systems in development.

	**P-CARDIO Wired Recorder Holder** The shielded wire ECG snap recorder holder can be used in combination with standard gel electrodes (or alternatively single dry electrodes). The recorder can be attached to the body with a lanyard or can be directly attached to the skin with a sticker. - **HPH231**: Wired 1-Lead + Respiration recorder holder - **HPH232**: Wired 6-Lead recorder holder - **HPH233**: Wired 6-Lead recorder holder – long wires
	**P-CARDIO Wireless BLE USB Dongle** **Connector**: USB Type A to computer **Requirements**: USB2.0
	**P-CARDIO Docking Station** **Connector:** mini-USB to computer **Universal connector**: connect to Recorder **Requirements**: USB2.0
	**P-CARDIO Software Application** **Minimal system requirements:** Windows 10+ 64bit / MAC, at least one free USB port

2.3 P-CARDIO Recorder



Light indicators

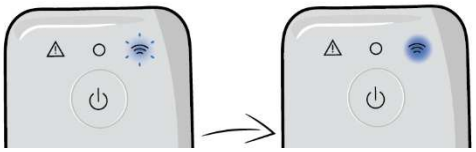
Press button for 1 to 3 sec	**(1) Shows device status:** Device is recording (solid green)  Device is not recording (solid orange) 
	(2) Starts advertising or shows wireless connection: Start advertising: (blue blinking)  Wireless BLE connected: (solid blue) 
	(3) If enabled, this also marks an event.
	Note: No light indicators: battery is empty.
Press button > 3 sec	**Start/stop if this option is enabled via the software**
	Starts the recording (fast orange blinking, followed by solid green after release) 
	Stops the recording (fast green blinking, followed by solid orange after release) 
Press button > 8 sec	**Device resets, then it will start advertising** (all LEDs blinking) 

Marking:

- Button marker: press button <3 sec
- Tapping marker using accelerometer: single tap while not moving

Connectivity indicator:

- Start advertising: blue blinking
- Wireless BLE connected: solid blue for 1min



Warning indicator:

Indication of warning that requires operator/ intervention.

Main warnings in case of:

- Lead-off (during recording): 2 orange blinks every 30 seconds
- Battery low: 1 orange blink with 10 sec intervals
- Memory full: 3 orange blinks with 10 sec intervals



Note: *Connect the recorder to the P-CARDIO Software Application to see more detailed information*

Charging:

For charging the recorder must be connected to docking station.

- charging in progress: slow blinking green
- charging completed: solid green

Turning on/off:

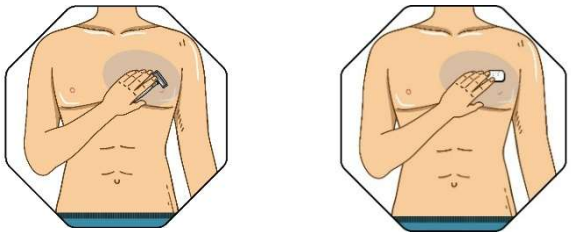
- Pressing the button will turn on the device.
- It turns off automatically after 5 minutes of no activity, if there is no scheduled recording within a week.

Universal Connector:

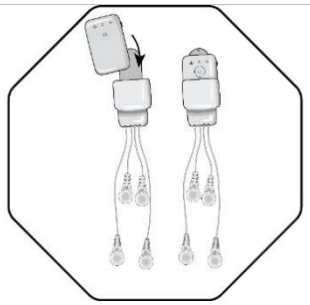
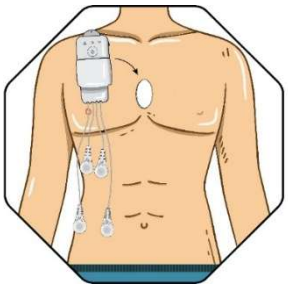
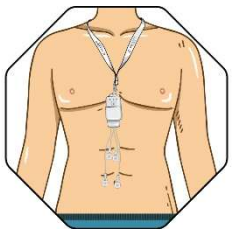
Universal connector port is used for charging and measurement setup & download.
Important: *Universal connector is non-standard and should only be used within the P-CARDIO eco system.*

2.4 Attachment of the PCARDIO System

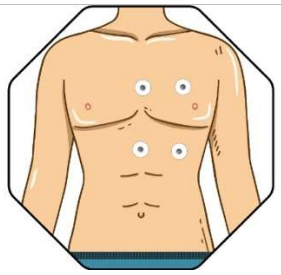
Prepare your skin

- If hair is on your chest in the grey area, shave using a razor. - Wash the area with soap and water. - Dry thoroughly with a towel. - Do not apply lotions or oils. - Once the skin is dry, scrub the area for 60 seconds with disinfect.	

Insert recorder in recorder holder

Fully slide the recorder in the recorder holder.	
<u>Option 1: Sticker</u> Remove the backing of the sticker for attaching the wired recorder holder to the body. Place the sticker firmly on your chest and attach the wired recorder holder to the sticker.	 
<u>Option 2: Lanyard</u> Click the recorder holder into the lanyard. Put the lanyard around your neck and adjust to your desired length.	 

Place ECG Electrodes

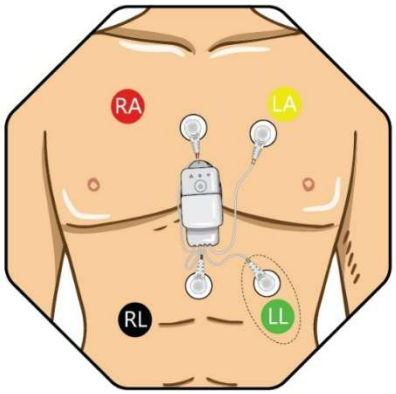
Remove the backing of the ECG electrodes.	
Place ECG electrodes at the chest, see illustration for electrode locations.	

Attach the snaps

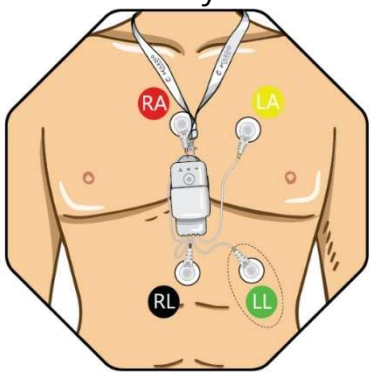
Attach the wired snaps to the ECG electrode stickers on your chest.
Place it based on the use case.
Refer to Einthoven triangle placement for optimal ECG signals.

One possible placement is shown below:

Option 1: with sticker



Option 2: with Lanyard



Color	Electrode	Explanation
Red	RA	Right Arm
Yellow	LA	Left Arm
Black	RL	Right Leg
Green	LL*	Left Leg

***** LL Green Electrode is not used for the 1-Lead recorder system.

2.5 P-CARDIO Software Application

P-CARDIO requires a computer running Windows 10 64 bits with at least one USB2.0 port or Apple (MacOS).

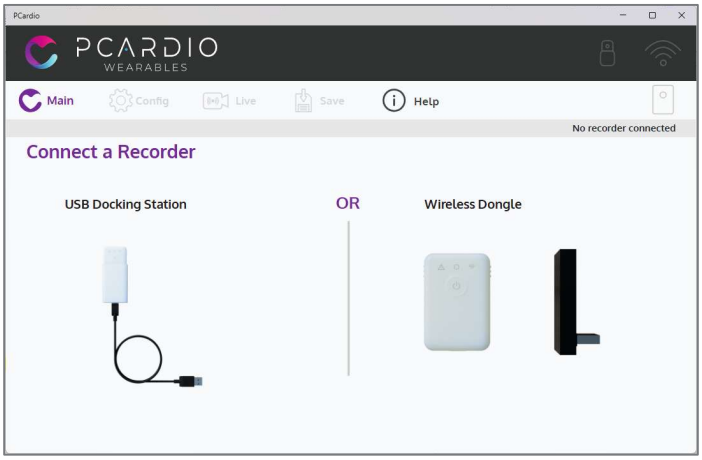
You can install the Software Application from the Microsoft or Apple App Stores.

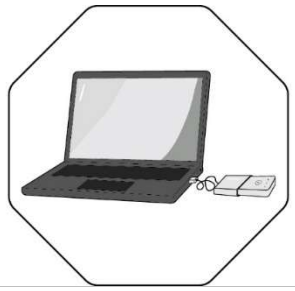
The computer automatically recognizes the Wireless BLE Dongle and the Docking Station.

Note: *You can check in the device manager if there is* ***2M BLE dongle*** *or PCARDIO CDC device present.*

Overview

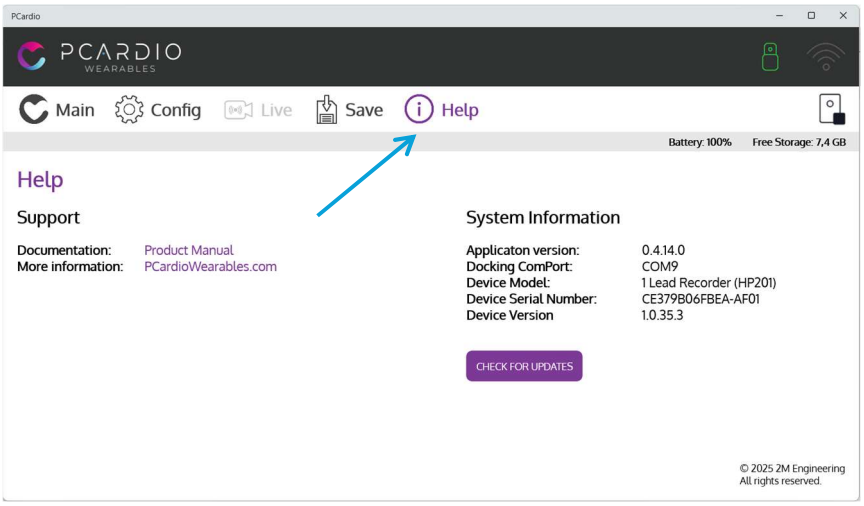
You can communicate with the Recorder via (1) Docked Connection or (2) Wireless Connection.



Docked Connection	Wireless Connection
	
Function: - Configuring device - Saving recording file	**Function:** - Live View - Start recording
Connection: - Connect the Docking Station to the computer via the USB cable. - Insert the Recorder to the Docking Station. Note: The Recorder also starts charging.	**Connection:** - Connect the dongle to the computer via the USB port. - Press Recorder button to connect.

System Information

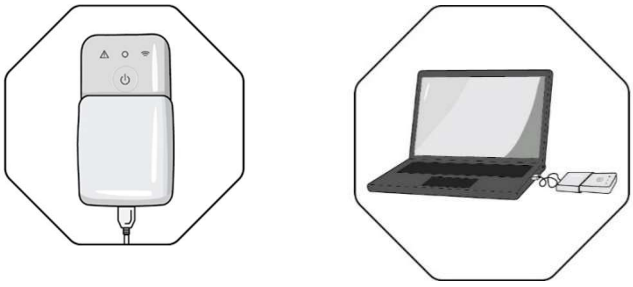
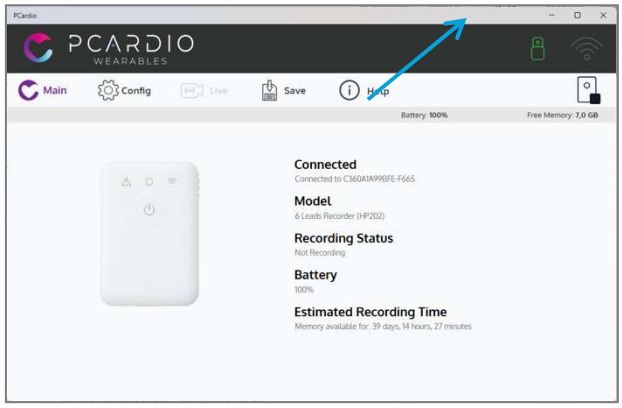
System Information can be accessed via the “Help” menu.



3 Using the P-CARDIO System

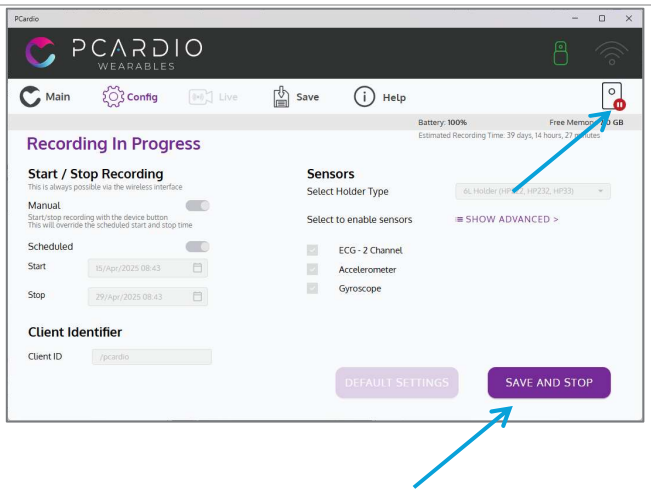
1) Before using the P-CARDIO system, make sure
   - the recorder is clean. See section 5 for cleaning instructions
   - the recorder is charged. See section 4 for charging instructions.
2) Configure the Recorder. See section 3.1.
3) Attach the System. See section 2.4.
4) Confirm the ECG Recording via Live View. See section 3.2.
5) Start/stop the Recording. See section 3.3.
6) Save the recording file. See section 3.4.

3.1 Configuring the Recorder

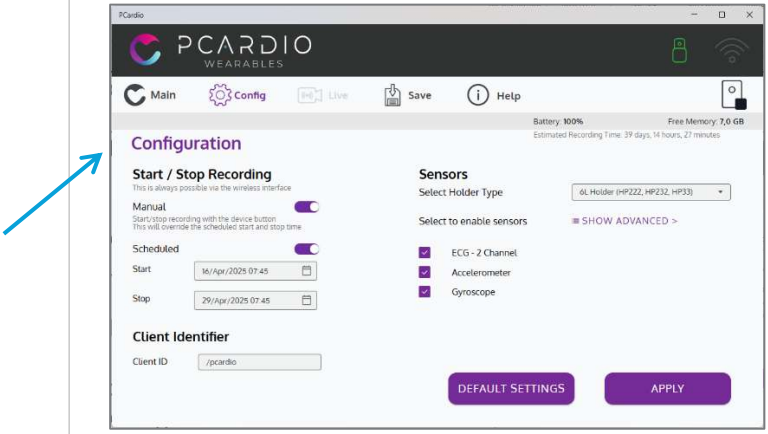
1. Use the Docked Connection to configure the recorder. - Start the P-CARDIO PC Application. - Connect the Docking Station to the computer via the USB cable. - Insert the Recorder to the Docking Station. Note: The Recorder also starts charging.	
2. The top bar will show the Docking connection. The "Main" menu will show device and recording information. - Connected: Recorder Serial Number - Model: Recorder Model - Recording Status - Battery level - Estimated Recording Time, based on available memory	
Note: If the device was recording, the recording will be paused.	

It is not possible to change the configuration during this state.

It is possible to stop recording in the “Config” menu by pressing the “Stop” button.

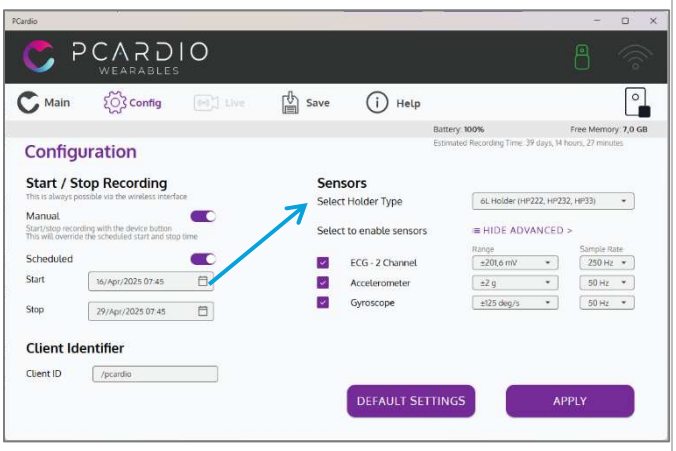


3. Select the “Config” menu.



4. **Configuring the Sensors**

- Select the type of the Recorder-holder you will be using. Available sensors depend on the type of the Recorder-holder.
- Enable the sensors by clicking the checkbox.
- Select the range and the sample rate for the sensor in the “Advanced” settings.
- Press “Default Settings” button to change the sensor settings to default values.
- Press “Apply” button to save the settings.



Estimated Recording Time is shown for the selected sensor configuration, on the top right corner. This value is based on the available memory.

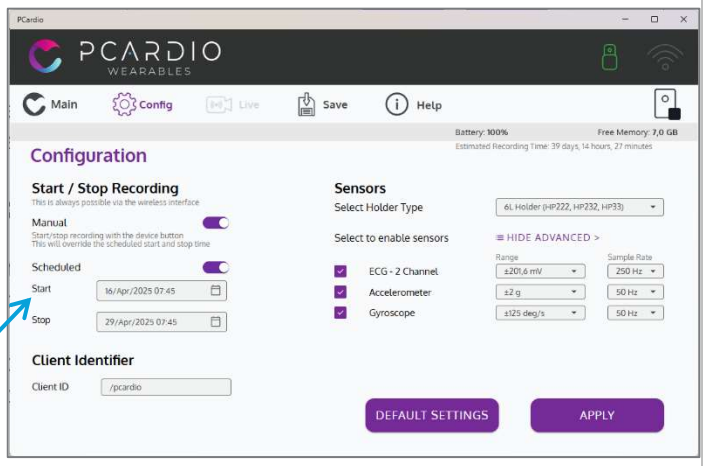
5. **Configuring the Start/Stop Time**

- Start/Stop is always possible via the Wireless Interface. Refer to section 3.3.

- Manual Start/Stop can be enabled via the “Manual” toggle bar.
  When Manual Start/Stop is enabled, pressing the Recorder button for 3 sec, starts and stops the recording.
  When Manual Start/Stop is enabled, starting/stopping with the Recorder button overrides the scheduled start/stop time.

- Start/Stop times can be scheduled via the “Scheduled” toggle bar.

- Press “Apply” button to save the settings.

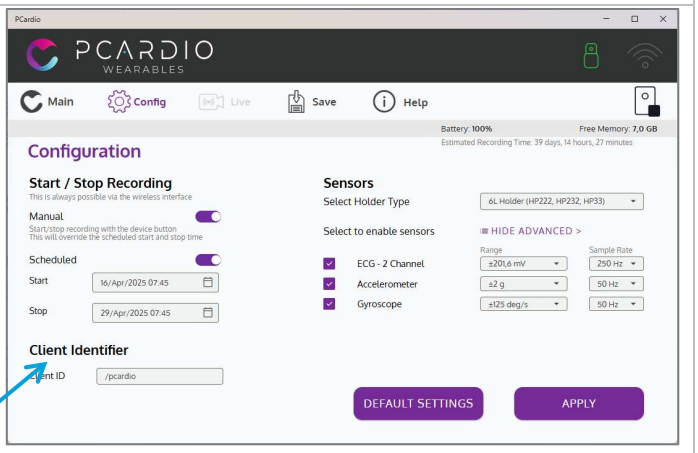


6. **Client Identifier**

Enter the Client Identifier to “Client ID”. (optional)
This field can be left empty or filled with up to 12 characters.

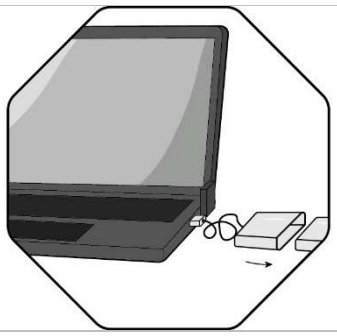
Note: The “Client ID” can also be entered when starting the recorder via the wireless connection. Refer to section 3.3.

Press “Apply” button to save the settings.



7. The recorder can be removed from the docking station.

Note: Always safely remove the docking station from your computer via the taskbar to avoid data corruption or damage to the device.

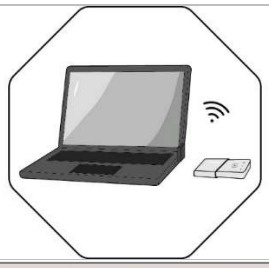


3.2 Confirm the ECG Recording via Live View in Wireless Connection

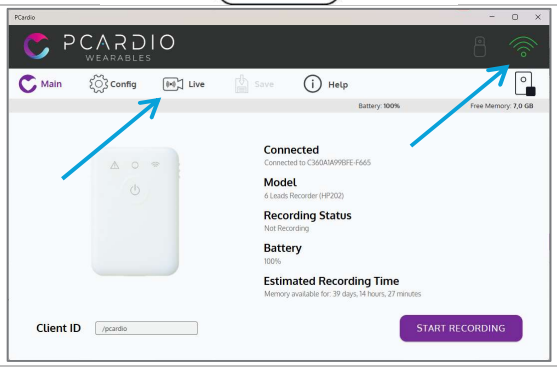
It is recommended to confirm the ECG Recording via Live View after attaching the system.

1. Initiate wireless connection

- Start the PCARDIO PC Application.
- Connect the dongle to the computer.
- Press button on the recorder to connect.



2. The top bar will show the Wireless connection.
3. Select the “Live View” menu



4. “Live View” shows:

- Live data from the enabled sensors
- “Lead Off” signal if detected

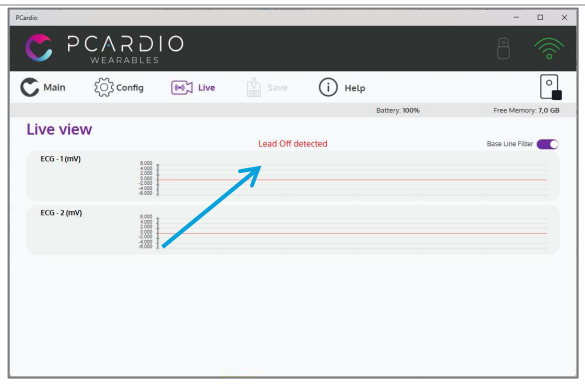
Base Line filter can be applied for visualization purposes.

This can be used for a quick check to see if the system has been applied correctly.



In case, there is Lead-off warning when measuring ECG, check the connections.
Make sure

- the recorder is inserted correctly in the recorder holder
- the connection between the electrodes and the recorder holder is correct
- the electrodes make a proper connection with the body.



3.3 Start/Stop

Start/Stop is configured in the Docked connection. Refer to section 3.1.

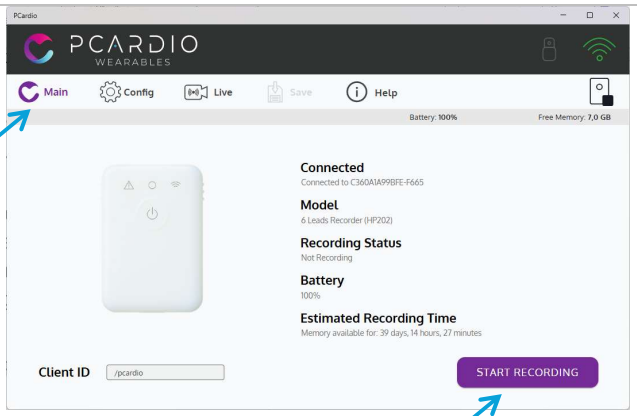
1. Start/Stop is always possible via the **Wireless connection.**

<u>Start recording</u>

a) In the wireless connection, select the "Main" menu.

b) Enter the Client Identifier to "Client ID". (optional)

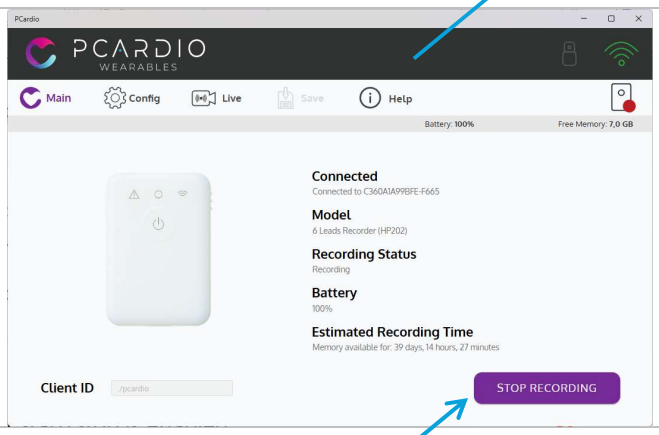
c) Press "Start Recording" button.



<u>Stop recording</u>

a) In the wireless connection, select the "Main" menu.

b) Press "Stop Recording" button.



2. When Manual Start/Stop is enabled:
   Press the Recorder button for >3 sec to start the recording.

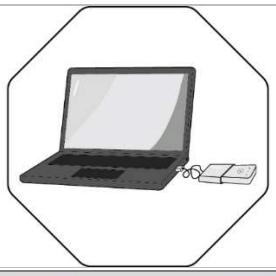
   Note: When Manual Start/Stop is enabled, starting/stopping with the Recorder button overrides the schedules start/stop time.

3. When Start/Stop is scheduled, recording will start/stop at the scheduled time. See section 3.1 for scheduling via docking.

3.4 Saving the Recording File

Use docking connection to save the recording data.

1. Connect the recorder to the computer via the docking station.

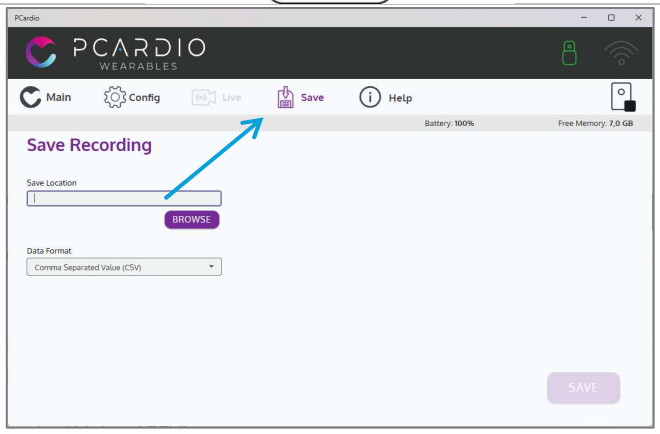


2. Select the “Save” menu.

3. Select the Location where the file will be saved.

4. Select the Data Format:
   Recording can be saved in two data formats:
   - Comma Separated Value (CSV)
   - European Data Format (EDF)
   - BioSemi Data Format (BDF)

5. Click on the “Save” button.



Recording File

For each recording, a recording folder is created with the start time as its name, “YYYYMMDDhhmmss”.

In this folder, there are recording files for each enabled sensor.

Note: Long recordings may be split into several folders.

CSV format

All recording files include a header row, including the sensor name and units for that specific sensor. The first two columns contain the date and time.

EDF format

All data is saved in one single file.

BDF format

All data is saved in one single file.

4 Charging the Recorder

Charge the Recorder using only (a) supplied USB-C cable or (b) the P-CARDIO Docking Station. The Recorder has a **non-standard** Universal connection.

Connect the recorder to (a) a computer or (b) a USB charger via the supplied USB-C cable or the PCARDIO docking station.

The indicator on the recorder is **blinking green** when charging and **solid green** when charging is completed, and the battery is full.

Note: Some computers switch off the charging power when powered off. Please check this to prevent having recorders that are not charged properly after a charging period.

Note: In case of using a USB charger, ensure that the output is +5Vdc and the current is at least 500mA.

Note: The charging time may vary depending on the current supplied by the USB-port. For a standard USB port with a current rating of max. 500mA, the charging time for a fully depleted battery is approx. 4-5 hours.

IMPORTANT: Make sure the Recorder is fully charged before use.

4.1 Battery and recording time

The recorder contains an internal Lithium polymer battery. The battery is built-in and cannot be replaced by the user. Internal safety circuitry protects the battery from overcharge and deep discharge.
Depending on the recorder type and settings, the recorder battery allows recording for at least 10 days.

5 Additional information

5.1 Cleaning

Wipe all external surfaces with 70% alcohol wipes.

5.2 Preventive inspection and maintenance

The P-CARDIO system does not contain any user serviceable parts.
There are no parts repairable by service personnel.
If there are malfunctions, send the device back to your supplier.

5.3 Troubleshooting

Connect the recorder to the docking station to get more information on the LED warnings on the recorder.

Contact 2M Engineering for further assistance. Please save the log file on the recorder. This will facilitate the troubleshooting process.

6 Technical Specifications

Exact specifications are still subject to change without further notice.

Function	Specifications	
Accelerometer	Range:	2/4/8/16G
	Dimensions:	X, Y, Z
	DC accuracy:	+/- 0,02G
	Sample rate:	50/100/200/400/800 Hz
	Resolution:	16bit
Gyroscope	Range:	125/250/500/1000 (default)/2000 degrees/sec
	Sample rate:	50/100/200/400/800 Hz
	Resolution:	16bit
Magnetometer	Sample rate	50 / 100 Hz
	Range	49.1 Gauss
ElectroCardioGraphy (ECG)	Channels:	1 Channel ECG or 2 Channels ECG
	DC common mode input:	0-3.0V
	Differential AC input:	2.5Vpp / 1.25Vpp / 625mVpp / 417mVpp / 313mVpp / 208mVpp
	Input Impedance:	10G Ω (without leadoff) 100-500M Ω (with leadoff)
	Input reference noise:	< 10µVpp
	Crosstalk channel separation:	> 50dB
	Filtering	EMI filter: 3MHz PGA RC-filter: 840Hz (anti-aliasing filter) Digital (On-Chip) filter: 65Hz (250sps), 130Hz (500sps), 260Hz (1000sps)
	CMRR:	>100dB
	Sample rates:	250/500/1000 sps
	Resolution:	24 bits with right leg drive
Respiration (Bioimpedance)	Sample rates:	250/500/1kHz
	Excitation frequency	32kHz / 64kHz
	Max Excitation current:	100 µA
	Resolution:	24 bits with right leg drive
Storage capacity	8Gb	
Measurement duration	ECG data at 250Hz: > 10 days ECG data at 500Hz: > 8 days (assumption: battery capacity to 80% over its lifetime)	
Event marker channel	The "ON" button is also used as marker channel. Sample rate: 1Hz.	
Battery monitoring	Internal continuous battery monitoring channel. Sample rate: 1Hz	
Battery	Internal rechargeable Lithium polymer battery. (3.7V, 400mAh nominal capacity)	
Dimensions and weight [L x H x D]	50 x 35 x 9 mm (recorder only) 20gr approx. 70 x 39 x 13 mm (recorder & holder). 50 grams approx..	
Recording duration	>5 days on one battery charge (depending on sample rate & configuration).	
Connectors	Non-standard Universal connector to connect to Docking Station.	
Docking station	Download speed	> 20Mbyte/sec
Environmental conditions		
Operation:	5°C to 45°C; 10% to 95% RH (Noncondensing)	
Storage:	0°C to 45°C; 10% to 95% RH (Noncondensing)	
Transport:	5% to 95% RH (Noncondensing)	
Connectivity	Wireless BLE 5.2 - Setup, start/stop, recording, live streaming	

7 Company info

PCARDIO Wearables is manufactured by 2M Engineering Ltd.



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